CITY UNIVERSITY MALAYSIA
Cyberjaya Campus — Faculty of Information Technology
BIT2083 Fundamental of Computational Thinking: Python
Final Project Report (40%)

15-Minute Neighborhood Score

A Beginner-Friendly Guide to a Python App (and Website) for SDG 11: Sustainable Cities and Communities

Full Name	[Your Full Name]
Student ID	[Your Student ID]
Class Code	[Your Class Code]
Program	[Your Program]
GitHub Repository	github.com/DevFadul/15-minute-neighborhood-score
Live Website	[paste your Render URL here once deployed]

August 2026

Note: replace the bracketed placeholders above with your own details (and each group member's) before submission, per the assignment instructions. NRIC/passport numbers are not included in this document — enter that information directly wherever your submission process requires it.

Contents

1. Introduction & SDG Background

Most people don't think twice about hopping in a car to buy milk or see a doctor. But that habit adds up: more traffic, more pollution, and neighborhoods that don't really work for anyone who can't drive — children, older adults, or families who can't afford a car.

This project is about United Nations Sustainable Development Goal 11 (SDG 11): **Sustainable Cities and Communities**. It's the UN's goal of making cities safer, greener, and easier to live in without needing a car for every errand. Two parts of that goal matter most here:

- **Target 11.2** — everyone should have access to safe, affordable transport.
- **Target 11.7** — everyone should have access to green space and public places.

Both are really about one simple question: *how close are the things you need to your front door?*

City planners have a name for a neighborhood that answers “close enough”: a **15-minute city**. The idea is that a resident should be able to walk to most of their daily needs — food, healthcare, school, public transit, parks, and everyday shops — in about 15 minutes.

This project turns that idea into something anyone can actually use. It's called the **15-Minute Neighborhood Score**, and it now exists in two forms: a Python console program, and — added later — a full interactive website with a 3D map you can explore. Both let a person score how walkable their own neighborhood is, and both tell them exactly what's missing.

2. Problem Statement

It's genuinely hard to know if a neighborhood is walkable just by living in it. You might know your nearest grocery store is close — but is the rest of the neighborhood just as good? And if you're choosing between two apartments, how do you compare them fairly, instead of just going with a gut feeling?

The tools that already exist for this tend to fall into two camps: professional GIS software built for city planners (too complex for a regular renter), or websites that need a live internet connection, an account, and a paid mapping API (not something everyone has access to).

So the problem this project solves is simple: let anyone — using just a basic computer, or a web browser — describe how long it takes them to walk to their nearest grocery store, clinic, school, bus stop, park, and shop, and turn that into one clear, comparable score, plus specific tips on what to improve. That helps people make a more informed, less car-dependent choice about where to live.

3. System Objectives

The 15-Minute Neighborhood Score is built to:

- Let a user record a “neighborhood assessment” — how many minutes it takes to walk to six essential categories of places.
- Turn that into a single score from 0 to 100, with an easy-to-understand rating: Excellent, Good, Fair, or Poor.

- Save every assessment, so a user's history is still there the next time they open the program.
- Point out exactly which categories are weakest, with specific, realistic tips for each one.
- Let a user compare two saved assessments side by side, to help decide between two homes.
- Never crash on bad input — always explain the mistake in plain language and ask again.
- Show real computational thinking and clean, modular Python code throughout.
- **(Added later)** Offer all of the above through an easy, visual website — including automatically estimating walk times for any real address, without the user needing to measure anything themselves.

4. Application Design

This project actually has two versions that share the exact same brain:

- **The console app** — the original version. Runs in a terminal window with a text menu.
- **The web app** — added afterward. Runs in a normal browser, with a 3D globe on the homepage, a visual dashboard, and even a first-person 3D view where you can walk around a simplified version of your neighborhood.

Both versions call the exact same Python scoring code to do the math. That code lives in one shared package (`neighborhood_score/`), so the console app and the website will always agree on the score for the same data.

4.1 How the Console App Is Organized

The console app is split into small files, each with one clear job, so nothing gets tangled together:

- `ui.py` — everything that gets printed to the screen.
- `app.py` — the menu loop; decides what happens when a user picks an option.
- `models.py` — defines what a single “assessment” actually is (a class called `Assessment`).
- `scoring.py` — does the actual math (a class called `ScoreCalculator`).
- `recommendations.py` — figures out which categories are weak and picks a tip for each.
- `storage.py` — saves and loads the data file (JSON format) on disk.
- `validation.py` and `constants.py` — small shared helpers used by everything else.

Each file only “talks” to its direct neighbors. For example, the screen-printing code never reads or writes the data file directly, and the data-saving code never prints anything to the screen. That separation makes each piece easy to test and easy to change without breaking the others.

Figure 1 shows how the program actually runs, step by step — starting the program, showing the menu, and following one option (Create New Assessment) all the way through.

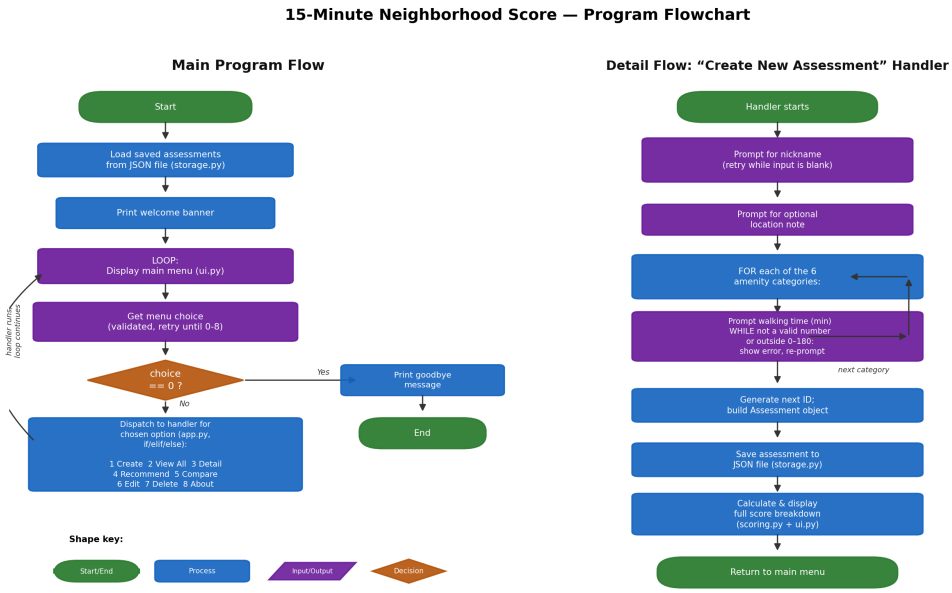


Figure 1. Program flowchart — main loop (left) and Create New Assessment detail flow (right).

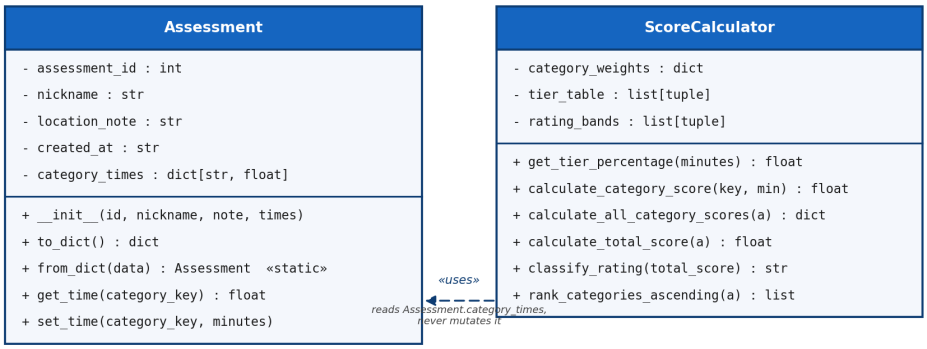
4.2 The Two Main Classes

Two classes sit at the center of the whole design:

- **Assessment** — just holds information about one neighborhood (its name and its six walking times). It doesn't do any math itself.
- **ScoreCalculator** — does the math. It takes an Assessment and turns it into a score, a rating, and a list of the weakest categories.

Keeping “the data” separate from “the math” is a common, useful pattern: it means the math can be tested on its own, and the data never accidentally gets out of sync with a stale, previously calculated score.

UML Class Diagram — 15-Minute Neighborhood Score



Design notes:

- Two classes only, each earning its place independently — data (Assessment) is kept separate from behavior (ScoreCalculator) to avoid one “God object” doing both jobs.
- Assessment never caches a total score or rating — both are always re-derived by ScoreCalculator on demand, so there is exactly one source of truth (no risk of a cached score going stale after an edit).
- All other application logic (validation, storage/persistence, recommendations, console presentation, menu control flow) lives in plain, single-purpose module-level functions in validation.py, storage.py, recommendations.py, ui.py and app.py — see the Implementation Details section for the full function inventory.

Figure 2. Class diagram for Assessment and ScoreCalculator.

4.3 How the Web App Is Built

The website reuses the exact `Assessment` and `ScoreCalculator` classes shown above — it just wraps a much friendlier interface around them, built with a Python web framework called Flask. Its main screens are:

- A rotating 3D globe on the homepage (Figure 3), where a user can type any real address.
- A real address lookup: the app looks up that address using free map data from OpenStreetMap, automatically finds the closest real grocery store, clinic, school, bus stop, park, and shop, and estimates realistic walking times — no manual measuring needed.
- A dashboard (Figure 4) listing every saved assessment as a card, each with a small map-style preview and a walkability “ring.”
- A first-person, 3D “walkthrough” screen where a user can literally walk around a simplified 3D neighborhood and watch the distance to each place update live.
- Simple forms, a detail page, and a compare page, all styled to look like pages from an old paper map or atlas.

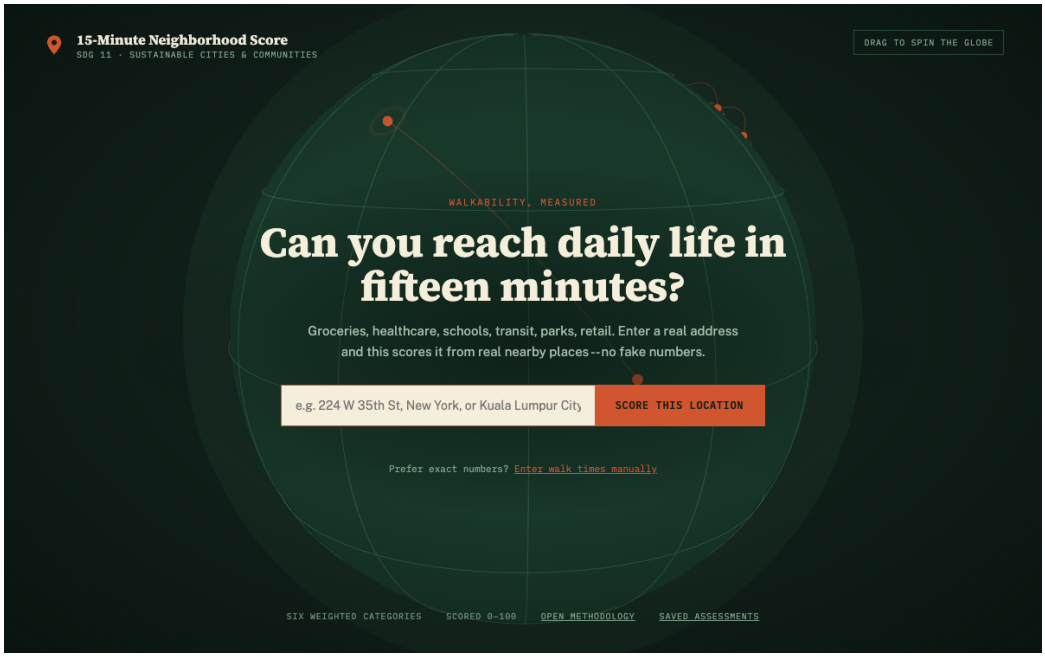


Figure 3. The web app's homepage — a rotating 3D globe with a real-address search box.

Figure 4. The web app's dashboard, listing saved assessments as cards.

5. Implementation Details

5.1 The Scoring Formula

Each of the six categories — Grocery, Healthcare, Education, Transit, Parks, and Retail — is worth a set number of points out of 100 (its “weight”). How many of those points a neighborhood actually earns depends on how close that category is:

Walking time	Points earned
0–5 minutes	100% of that category's points
6–10 minutes	75%
11–15 minutes	50%
16–25 minutes	20%
More than 25 minutes	0%

Add up the points earned across all six categories, and the result is a single score from 0 to 100. That score is then turned into a simple rating: **Excellent** (85 or above), **Good** (70–84), **Fair** (50–69), or **Poor** (below 50).

5.2 Estimating Real Addresses (Web App Only)

When a user types a real address into the website, the app does three things automatically:

- Turns the address into map coordinates (latitude and longitude), using a free service called **Nominatim**.
- Searches nearby, real-world map data (from a free service called **Overpass**) for the closest grocery store, clinic, school, bus stop, park, and shop.
- Estimates the walking time to each one, based on the straight-line distance and an average walking speed.

This isn't a perfect replacement for actually walking the route — a straight line “as the crow flies” is usually a bit shorter than the real path along streets and sidewalks — but it gives a genuinely useful, realistic estimate in seconds, for any real address, without the user needing to measure anything by hand.

5.3 Never Crashing on Bad Input

Every question the console app asks the user goes through the same small set of validation functions, so mistakes are always handled the same way. If someone types letters instead of a number, a negative number, or an unrealistically large travel time, the program doesn't crash — it explains exactly what was wrong and asks again. The same care applies to saved data: if the data file on disk somehow gets corrupted, the program doesn't crash on startup either — it renames the broken file, warns the user, and simply starts with an empty history instead.

6. Testing & Sample Outputs

6.1 How It Was Tested

The console app was tested by actually running it and feeding it real input — including a lot of deliberately wrong input — and checking that it always responded correctly instead of crashing. This included:

- Starting the program with no saved data yet, and with a corrupted data file.
- Typing invalid menu choices, invalid numbers, negative numbers, and unrealistically large numbers.
- Checking the score and rating at every boundary of the walking-time table (e.g., exactly 5 minutes vs. 6 minutes).
- Getting recommendations for both a perfect score and a very low score.
- Comparing two different assessments, and trying (and correctly failing) to compare an assessment against itself.
- Editing and deleting assessments, then confirming the changes were still there after closing and reopening the program.

All of these worked as expected. A short example is shown below — this is real output from the program, not a mockup:

```
Enter a nickname for this neighborhood: Current Apartment
Enter an optional location note (or press Enter to skip): Jalan Ampang, KL

Grocery / Fresh Food Access (minutes): 4
Healthcare / Pharmacy (minutes): 12
Education / Childcare (minutes): 20

[OK] Assessment #1 "Current Apartment" saved.
------------------------------------------------------------
SCORE BREAKDOWN: Current Apartment  (ID 1)
------------------------------------------------------------
Category                        Minutes   Tier      Points
------------------------------------------------------------
Grocery / Fresh Food Access         4.0   100%   20.0 / 20
Healthcare / Pharmacy              12.0    50%    7.5 / 15
Education / Childcare              20.0    20%    3.0 / 15
------------------------------------------------------------
TOTAL SCORE: 73.0 / 100   ->   Rating: GOOD
```

Excerpt from tests/manual_test_transcript.txt (shortened here for space).

The web app was tested by hand in a real browser: creating, editing, and deleting assessments; watching the live score preview update while filling in the form; comparing two assessments; walking through the 3D neighborhood view; and searching several real addresses — from a dense city center to a remote location — to confirm the automatic walk-time estimate behaves sensibly in both cases.

6.2 A Real Bug Found During Testing

Testing wasn't just a formality — it found a real bug. In an early version of the “Compare Two Assessments” screen, a long neighborhood nickname could overflow its column in the results table and run into the next column, because Python doesn't automatically shorten long text to fit a fixed width. The fix added a small helper function that safely shortens any text that's too long for its space, and applied it everywhere a name gets displayed — not just where the bug was first noticed.

7. Discussion & Limitations

By turning “is my neighborhood walkable?” into one clear, comparable number, this project gives people a genuinely practical tool for making a more sustainable choice about where to live — which is exactly what SDG 11 is about. The comparison feature addresses a real decision people actually face (choosing between two homes), and the recommendations turn a plain number into something actionable.

A few honest limitations remain:

- The automatic address lookup (in the web app) estimates walking time using a straight line, not the actual walking route along real streets — so it's a helpful estimate, not an exact measurement.
- Every user gets the same category weights, even though different people have different priorities — a family with young children might care about Education far more than a retired couple would.
- The score measures whether something is nearby in time, not whether it's safe, pleasant, or well-lit to actually walk to.
- Data is stored in one file (or, once the site is deployed online, in the hosting service's storage) rather than a proper shared database — enough for one person's own use, but not built for many people using it as a shared service at once.

Worth noting: one limitation from the original version of this project — “no mapping integration, so the user has to estimate their own walking times” — has since been solved. The web app now does exactly that automatically, for any real address, using free map data.

8. Conclusion

This project set out to use Python and computational thinking to solve a real, everyday sustainability problem under SDG 11: helping people understand and compare how walkable their neighborhood actually is. The result is now two working tools that share one brain: a tested, menu-driven console program, and a full interactive website — complete with a 3D globe, real address lookups, and a 3D neighborhood you can walk around — that turns a vague feeling of “this area seems walkable” into a clear, useful, and comparable score.